

PHYS598

INVESTIGATING TWO-STEP PERFECT QUANTUM STATE TRANSFER IN QUANTUM WALKS

Nathan Ngo

Supervisor: David Feder

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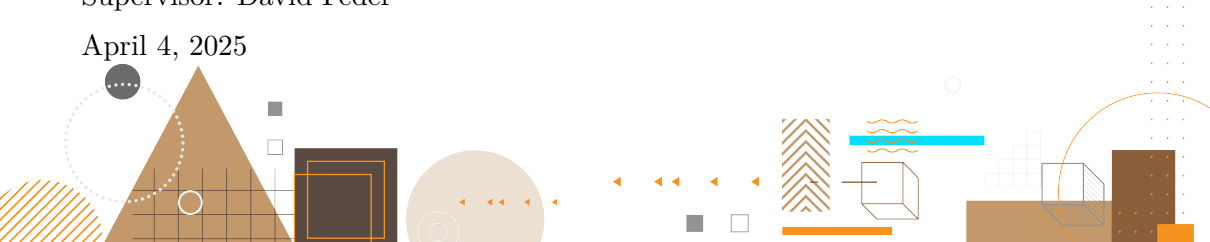


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1 Classical Random Walk

- ▶ Classical Random Walk
- ▶ Quantum Walk
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- ▶ One-Step Perfect Quantum State Transfer (PST)
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- ▶ Research Questions
- ▶ Methodology
- ▶ Results
- ▶ Conclusion
- ▶ Future Directions

What is it?

1 Classical Random Walk

- Moves step by step in a certain space, according to specific probabilistic rules

Discrete-Time Classical Random Walk

1 Classical Random Walk

- **Discrete time:** Moves at specific, regular intervals



Continuous-Time Classical Random Walk

1 Classical Random Walk

- **Continuous time:** Moves at any time, not just at fixed intervals

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2 Quantum Walk

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What is it?

2 Quantum Walk

- In quantum mechanics, particles can exist in superpositions
- Particles can be in a superposition of several positions simultaneously
- Quantum interference¹

¹cite-key.

Continuous-Time Quantum Walk

2 Quantum Walk

- Evolves smoothly rather than in discrete steps
- **Rules:** Governed by the time-dependent Schrodinger equation

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3 Graph Theory

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What is it?

3 Graph Theory

- Mathematical study of structures called graphs: composed of nodes (vertices) and edges (connections)
- Graphs model relationships between objects in a variety of fields

How is a Quantum Problem also a Graph Problem?

3 Graph Theory

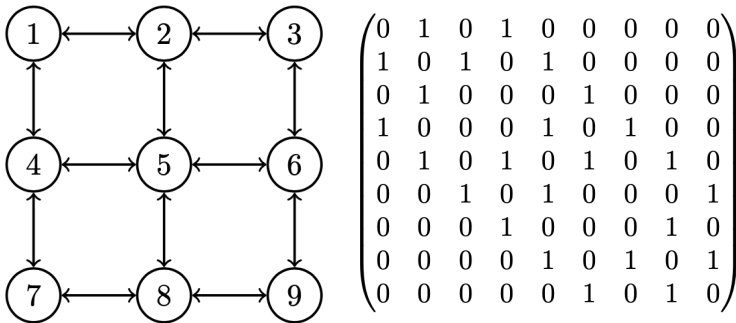


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4 One-Step Perfect Quantum State Transfer (PST)

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What is it?

4 One-Step Perfect Quantum State Transfer (PST)

- A rare phenomenon²
- An exact transfer of a quantum state from node A to node B with a perfect fidelity³
- A promising tool for quantum information science²

²COUTINHO2022155.

³BabukhinD.V.2022Teoq.

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5 Two-Step Perfect Quantum State Transfer (PST)

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Differences between One-Step and Two Step PST

5 Two-Step Perfect Quantum State Transfer (PST)

- Let the quantum state evolve on the initial graph configuration for a specified duration
- Introduce a controlled modification to the graph structure
- Allow the system to evolve for a second period

Why do we care?

5 Two-Step Perfect Quantum State Transfer (PST)

- High flexibility
- Avoid decoherence

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6 Research Questions

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What is left to figure out?

6 Research Questions

- Which graph structures support two-step PST?
- What are the conditions for two-step PST?
- How do edge weights and vertex configurations affect PST?
- How to validate theoretical predictions?

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7 Methodology

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- ▶ **Methodology**
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How would I do it?

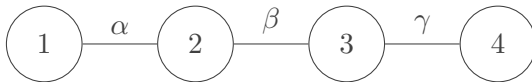
7 Methodology

1. Graph Selection
2. Optimizer implementation
3. Identification of mathematical structures
4. Analytical verification
5. Pattern recognition

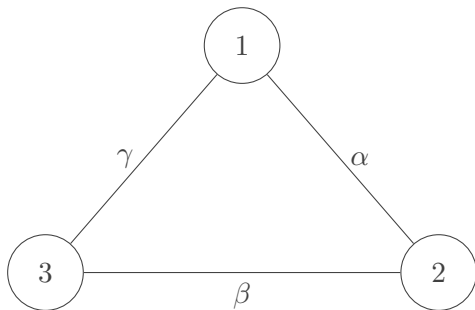
Graph Selection

7 Methodology

- P_4 :



- C_3 :

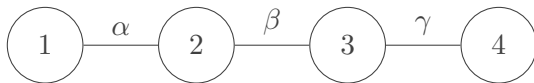


where $\alpha, \beta, \gamma \in \mathbb{R}$.

Why P_4 ?

7 Methodology

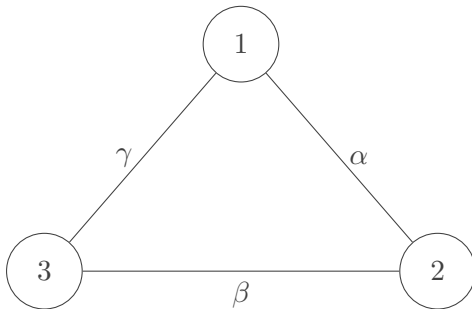
- Two known cases of two-step PST from node 1 to 4
- Motivates search for a broader family with similar behavior



Why C_3 ?

7 Methodology

- No known PST cases with real edge weights — all known cases use imaginary weights



Optimizer Implementation

7 Methodology

- Implemented in Python using TensorFlow
- Used Adam optimizer to maximize fidelity
- Variables: edge weights (before/after), and runtime (before/after adjustment)
- Output: Lists of irrational numbers

$$\text{Fidelity} \leq 0.999999999999$$

Initial Edge Weights	Final Edge Weights	$t = t'$
[1.0, -2.2677868380760757, 1.0]	[-1.0, -2.2677868380760757, -1.0]	2.077831382
[1.0, -2.2677868380760757, 1.0]	[-1.0, -2.2677868380760757, -1.0]	2.077831382
[1.0, -7.281358514018219, 1.0]	[-1.0, -7.281358514018219, -1.0]	5.824775131
[1.0, 0.1672484025560938, 1.0]	[-1.0, 0.1672484025560938, -1.0]	9.392512825

Identification of Mathematical Structures

7 Methodology

$$-2.2677868380760757 \rightarrow -\frac{6}{\sqrt{7}}$$

$$2.077831382 \rightarrow \frac{\sqrt{7}}{4}\pi$$

Analytical Verification

7 Methodology

- Substitute guessed mathematical forms into Mathematica
- Verify if fidelity evaluates to exactly 1

Pattern Recognition

7 Methodology

- Identify patterns in edge weights (before/after)
- Relate edge weights to corresponding runtimes
- Generalize to a broader family

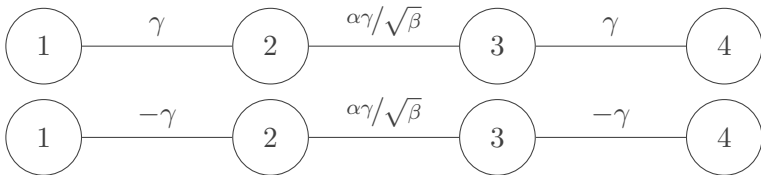
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P_4 : Two-Step PST from Node 1 to Node 4

8 Results



$$\beta = 7 + 8n = (5 + p)(3 - p + 8m), \quad (1)$$

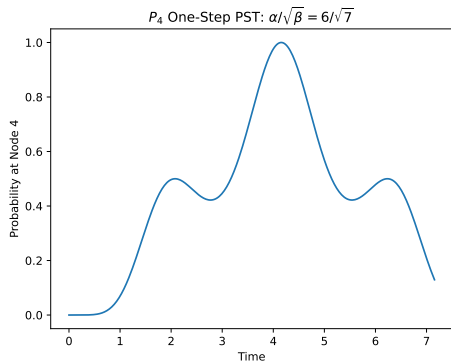
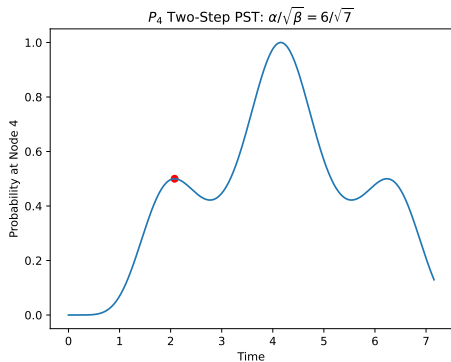
$$\alpha = 2 + 2p - 8m, \quad (2)$$

$$\text{where } \begin{cases} n = 0, 1, 2, \dots, \\ m = 0, 1, 2, \dots, n. \end{cases}$$

If $p \in \mathbb{Z}$ and $\alpha/\sqrt{\beta}$ is in its simplest form $\alpha'/\sqrt{\beta'}$ where $\alpha', \beta' \in \mathbb{Z}$, then the shortest time needed to achieve perfect fidelity is $t = t' = \frac{\sqrt{\beta'}\pi}{4\gamma}$.

Two-Step PST Revealed as One-Step PST

8 Results



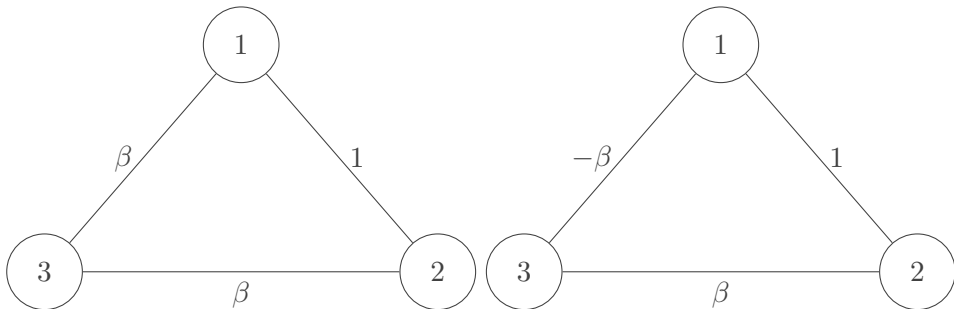
Two-Step PST Revealed as One-Step PST

8 Results

- Two-step PST family found to be realizable in a single step
- For P_4 , adjusting edge weight at a probability peak has no effect on dynamics
- Verified in Mathematica as a new one-step PST family

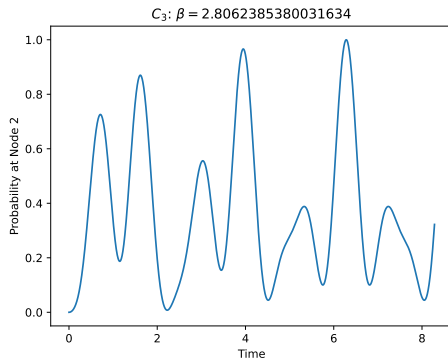
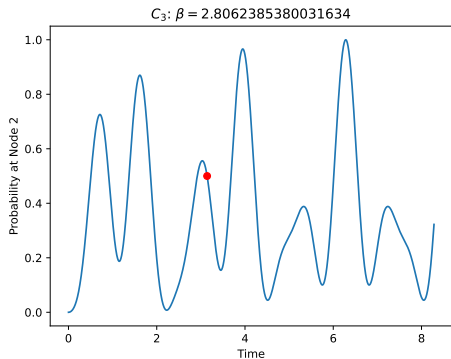
C_3 : Two Step PST from Node 1 to Node 2

8 Results



Honourable Mentions

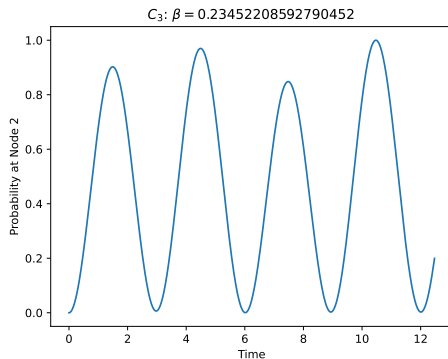
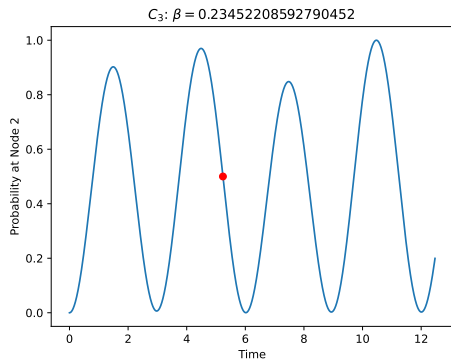
8 Results



Fidelity ≥ 0.999999999999 at $t \approx 2\pi$

Honourable Mentions

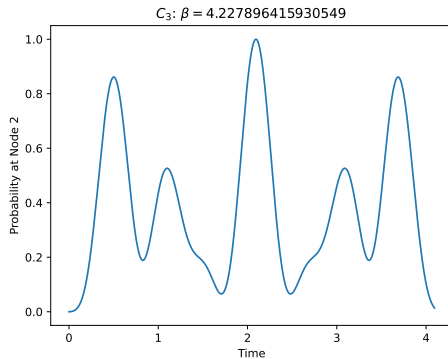
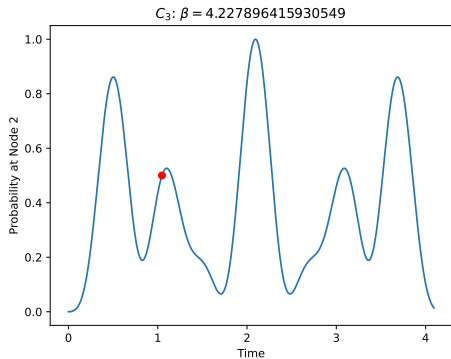
8 Results



Fidelity ≥ 0.999999999999 at $t \approx 10\pi/3$

Honourable Mentions

8 Results



Fidelity ≥ 0.999999999999 at $t \approx 2\pi/3$

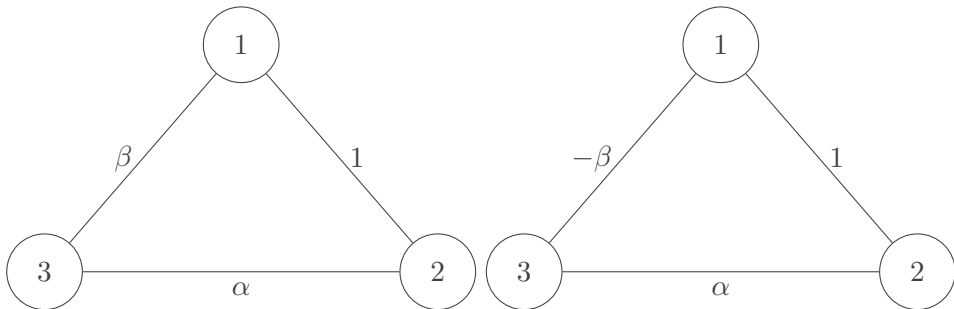
Is the Second Step Useless?

8 Results

- Two-step PST family can be achieved in a single step
- Timing of edge weight adjustment appears random, not aligned with probability peak
- Second step does not affect the system dynamics

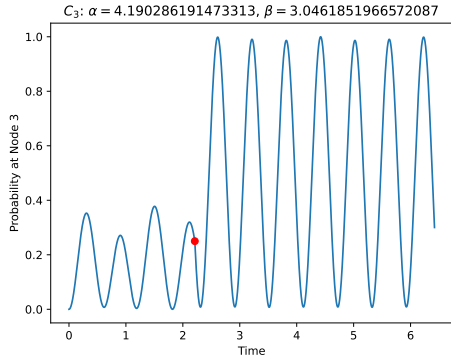
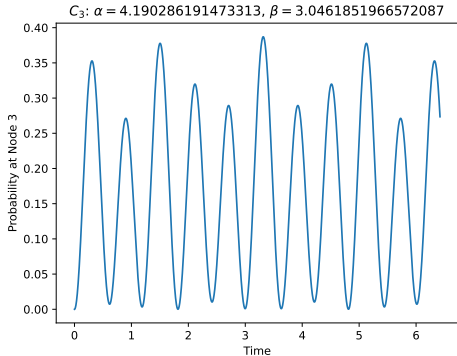
C_3 : Two-Step PST from Node 1 to Node 3

8 Results



When the Second Step Matters

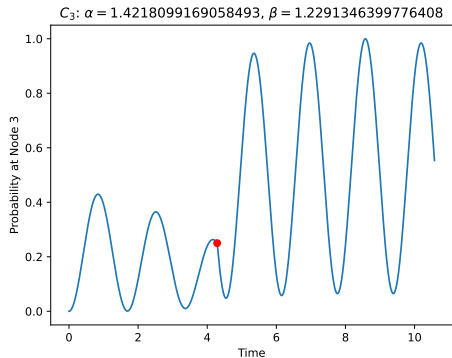
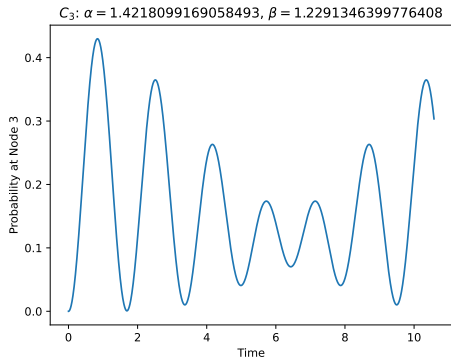
8 Results



Fidelity ≥ 0.999999999999 at $t = 4.420307096$

When the Second Step Matters

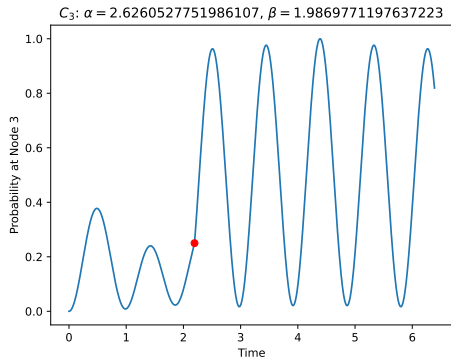
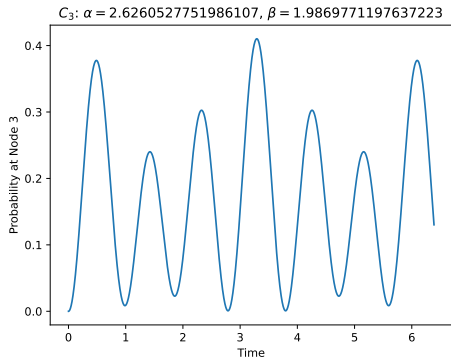
8 Results



Fidelity ≥ 0.999999999999 at $t = 8.576406656$

When the Second Step Matters

8 Results



Fidelity ≥ 0.999999999999 at $t = 4.38840648$

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Conclusion

9 Conclusion

- Extended PST on P_4 from two known cases to an unbounded family
- Real-weighted PST on C_3 appears likely, suggested by the π factor in t , but not yet confirmed
- One-step and two-step PST configurations are not isolated—evidence points to an underlying relationship

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Open Questions

10 Future Directions

- Can PST on path graphs P_n be characterized by a general rule?
- Is two-step PST the only mechanism enabling perfect fidelity on asymmetrically weighted graphs?
- Does allowing three-step PST reveal new or broader families?



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Q&A

Thank you for listening!
Your feedback will be highly appreciated!